

EES 201: Introduction to Earth Sciences

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Source: Department of Earth and Planetary Sciences –
Northwestern - Northwestern University

■ What is sedimentology

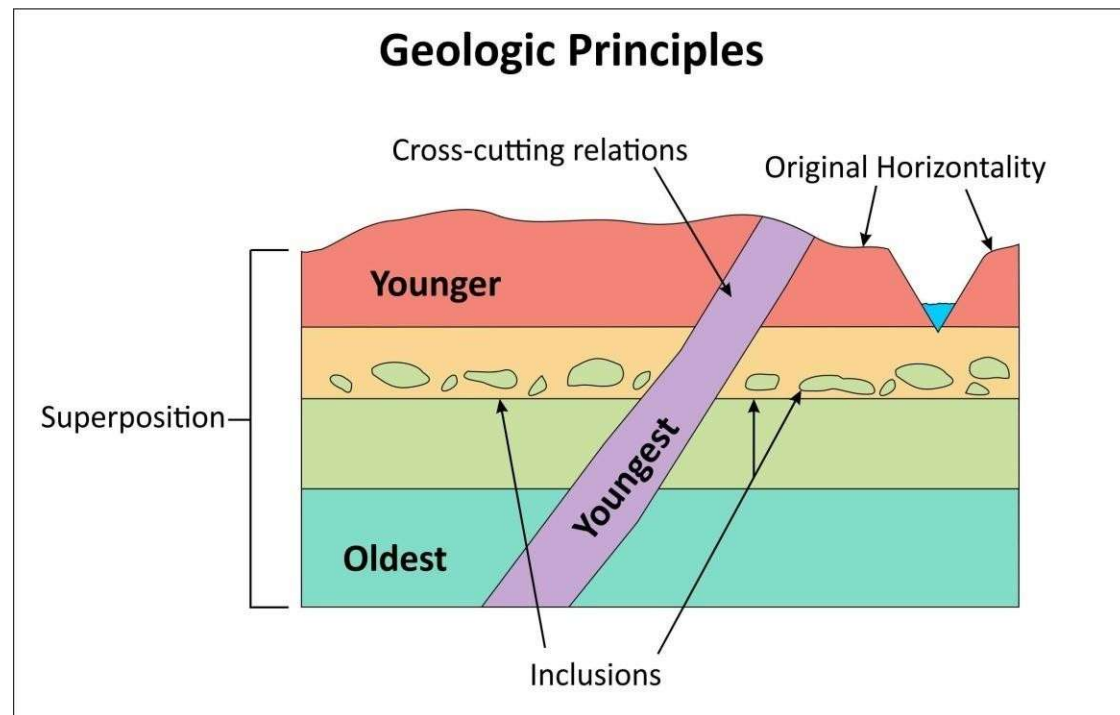
- Study of the processes of **formation**, **transport** and **deposition** of material that accumulates as sediment in continental and marine environments and eventually forms **sedimentary rocks** (G. Nichols).

Brief History

- The history of Sedimentology dated back to about 1500 AD.

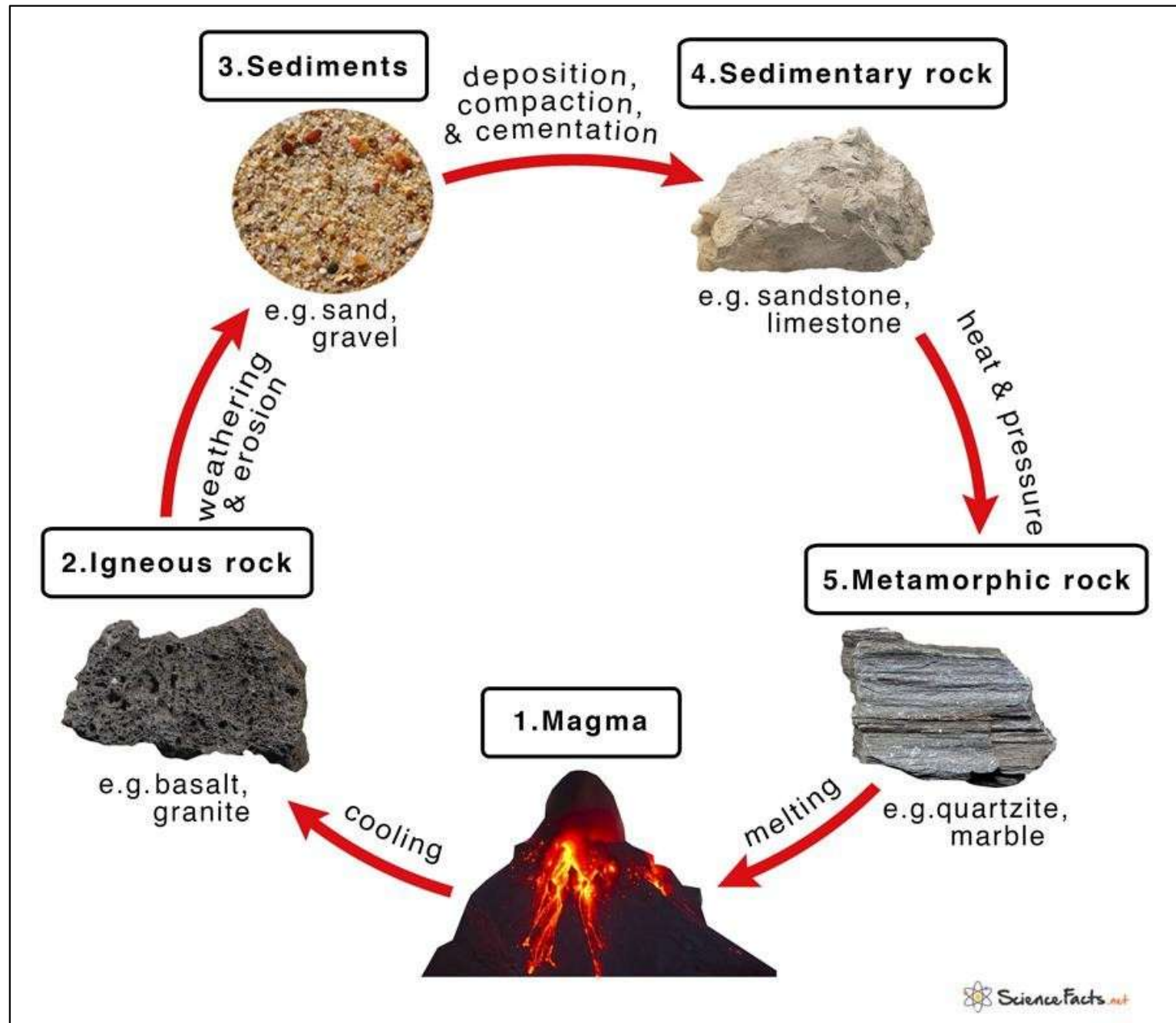
- Principle of superposition (Nicolas Steno, 1669).

- *“within a sequence of layers of sedimentary rock, the oldest layer is at the base and that the layers are progressively younger with ascending order in the sequence”*



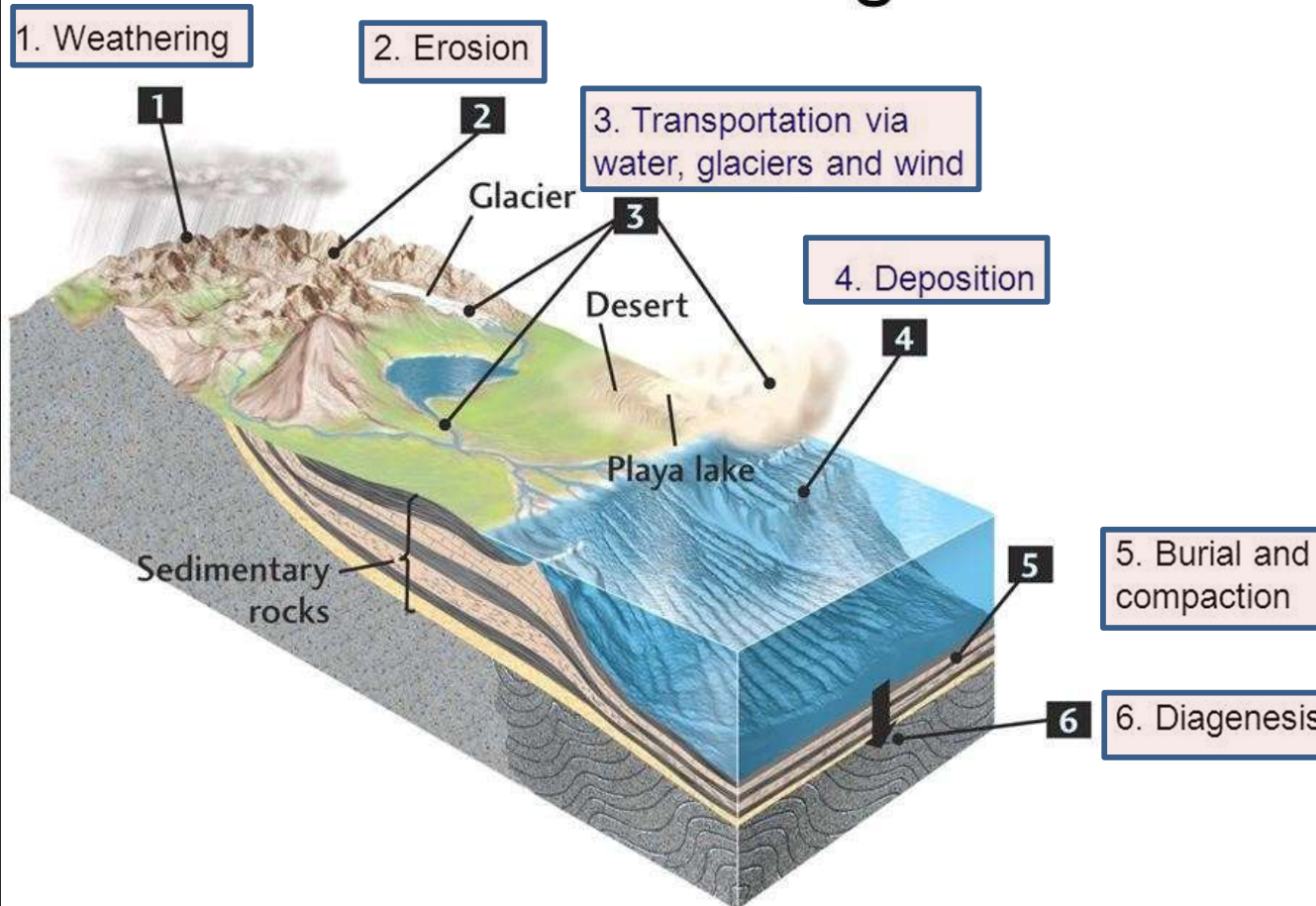
- Uniformitarianism (“the present is the key to the past”) (Charles Lyell, 1830): is the assumption that the same natural laws and processes that operate in our present-day scientific observations have always operated in the universe in the past and apply everywhere in the universe

Sedimentary rock



Sedimentary rock

Sediment stages



- Weathering & Erosion
 - Decomposition
 - Disintegration
- Transportation and deposition
- Diagenesis and lithification
- Compaction and cementation.

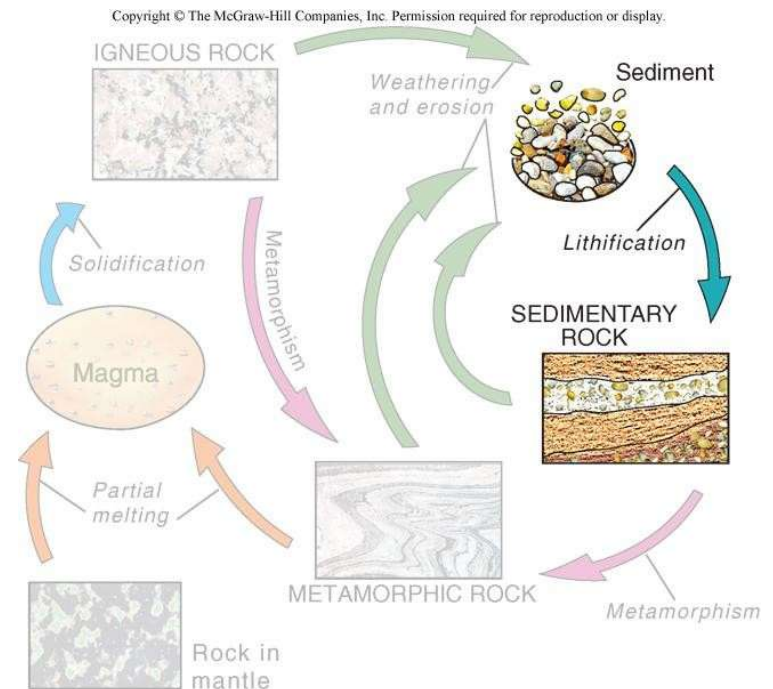
Introduction to Sedimentary rock

- *Sediment* - loose, solid particles originating from:
 - Weathering and erosion of pre-existing rocks
 - Chemical precipitation from solution, including secretion by organisms in water

- Classified by *particle size*

- Boulder - >256 mm
- Cobble - 64 to 256 mm
- Pebble - 2 to 64 mm
- Sand - 1/16 to 2 mm
- Silt - 1/256 to 1/16 mm
- Clay - <1/256 mm

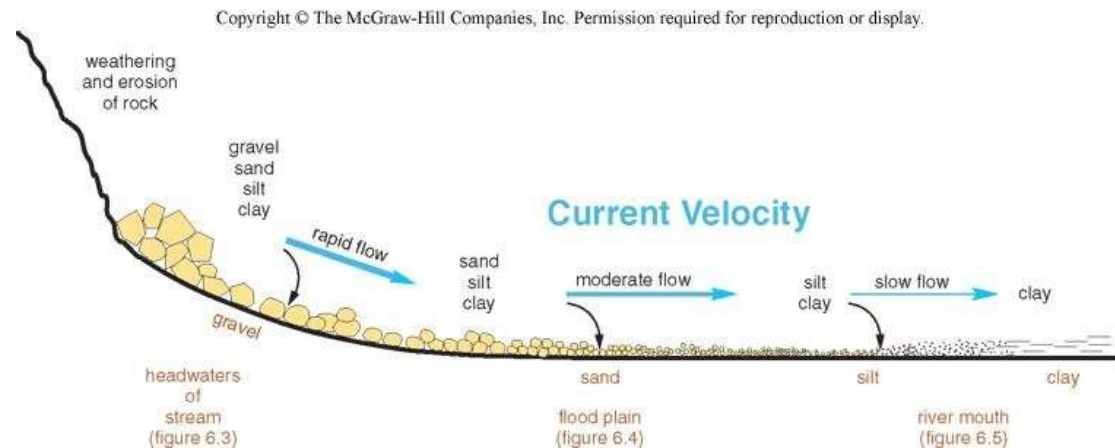
Gravel



Introduction to Sedimentary rock

- *Transportation*

- Movement of sediment away from its source, typically by water, wind, or ice
- *Rounding* of particles occurs due to abrasion during transport
- *Sorting* occurs as sediment is separated according to grain size by transport agents, especially running water
- Sediment size decreases with increased transport distance



Introduction to Sedimentary rock

- *Deposition*

- Settling and coming to rest of transported material
- Accumulation of chemical or organic sediments, typically in water
- *Environment of deposition* is the location in which deposition occurs
 - Deep sea floor
 - Beach
 - Desert dunes
 - River channel
 - Lake bottom



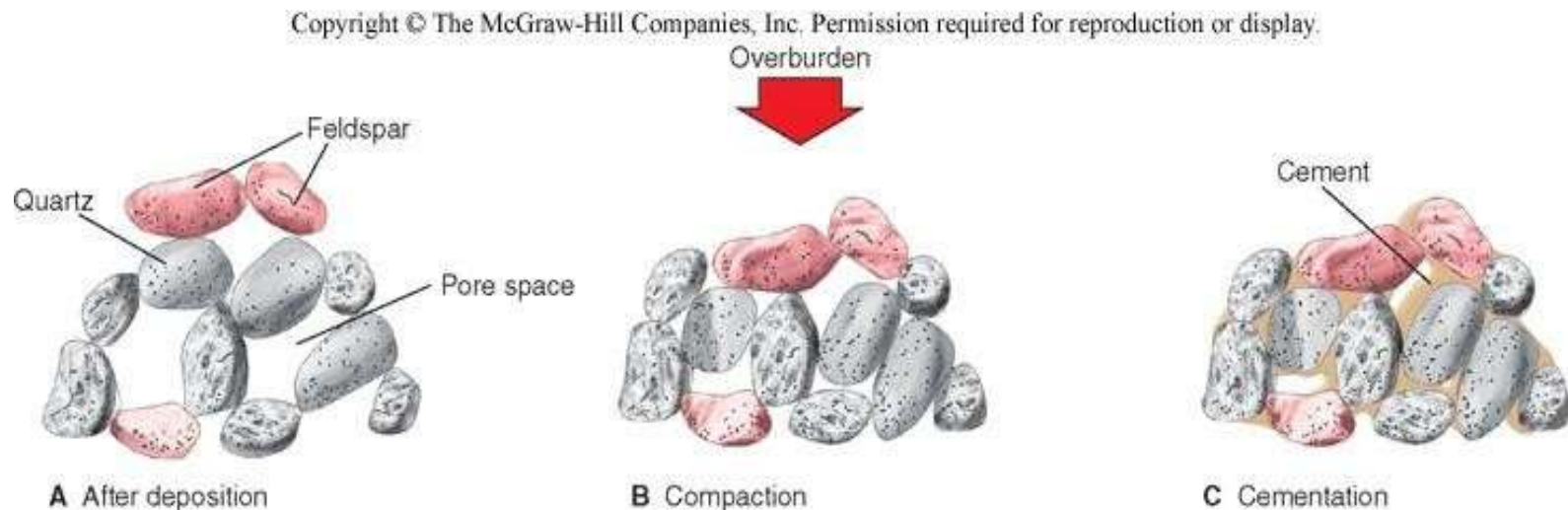
Introduction to Sedimentary rock

- *Preservation*

- Sediment must be preserved, as by burial with additional sediments, in order to become a sedimentary rock

- *Lithification*

- General term for processes converting loose sediment into sedimentary rock
- Combination of *compaction* and *cementation*



Classification of Sedimentary rock

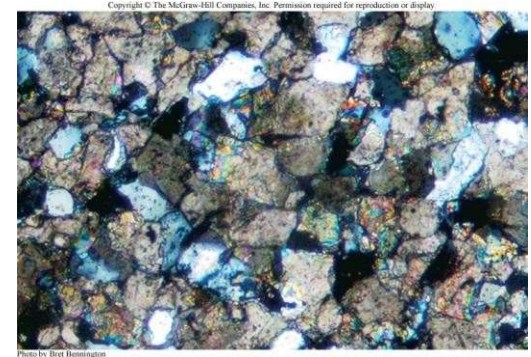
- *Detrital (clastic) sedimentary rocks*

- Most common sedimentary rock type
- Form from cemented sediment grains that come from pre-existing rocks



- *Chemical sedimentary rocks*

- Have crystalline textures
- Form by precipitation of minerals from solution



- *Organic sedimentary rocks*

- Accumulate from remains of organisms



Classification of Sedimentary rock: Clastic rocks

- *Breccia and Conglomerate*

- *Coarse-grained clastic* sedimentary rocks



- *Sandstone*

- *Medium-grained clastic* sedimentary rock
- Types determined by composition
 - *Quartz sandstone* - >90% quartz grains
 - *Arkose* - mostly feldspar and quartz grains
 - *Graywacke* - sand grains surrounded by dark, fine-grained matrix, often clay-rich



Classification of Sedimentary rock: Clastic rocks

- *Shale*

- Fine-grained clastic sedimentary rock
- Splits into thin layers (*fissile*)
- Silt- and clay-sized grains



- *Siltstone*

- Slightly coarser-grained than shales
- Lacks fissility

- *Claystone*

- Predominantly clay-sized grains; non-Fissile

- *Mudstone*

- Silt- and clay-sized grains; massive/blocky



Classification of Sedimentary rock: Chemical rocks

- *Carbonates*

- Contain CO_3 as part of their chemical composition
- *Limestone* is composed mainly of *calcite*
 - Most are *biochemical*, but can be *inorganic*
 - Often contain easily recognizable fossils
 - Chemical alteration of limestone in Mg-rich water solutions can produce *dolomite*



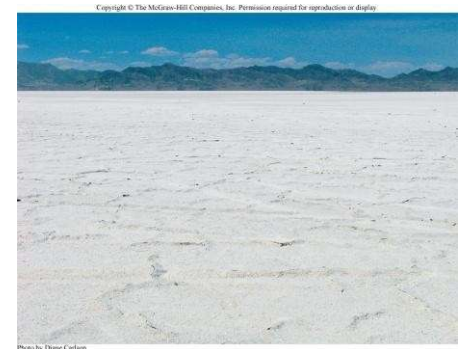
- *Chert*

- Hard, compact, fine-grained, formed almost entirely of silica
- Can occur as layers or as lumpy nodules within other sedimentary rocks, especially limestones



- *Evaporites*

- Form from evaporating saline waters (lake, ocean)
- Common examples are rock gypsum, rock salt



Classification of Sedimentary rock: Organic rocks

- Coal
 - Sedimentary rock forming from compaction of partially decayed plant material
 - Organic material deposited in water with low oxygen content (i.e., stagnant)
- Oil and natural gas
 - Originate from organic matter in marine sediment
 - Subsurface “cooking” can change organic solids to oil and natural gas
 - Can accumulate in porous overlying rocks

Sedimentary structures

- Features within sedimentary rocks produced during or just after sediment deposition
- Provide clues to how and where deposition of sediments occurred
- *Bedding*
 - Series of visible layers within a rock
 - Most common sedimentary structure
- *Cross-bedding*
 - Series of thin, inclined layers within a horizontal bed of rock
 - Common in sandstones
 - Indicative of deposition in ripples, bars, dunes, deltas



Sedimentary structures

- *Ripple marks*

- Small ridges formed on surface of sediment layer by moving wind or water



- *Graded bedding*

- Progressive change in grain size from bottom to top of a bed



- *Mud cracks*

- Polygonal cracks formed in drying mud



- *Fossils*

- Traces of plants or animals preserved in rock
- Hard parts (shells, bones) more easily preserved as fossils



Classification of Sedimentary rock

- Produced from *weathering products* of pre-existing rocks or accumulated *biological matter*
 - *Detrital* (clastic) rocks produced from rock fragments
 - *Chemical* rocks produced by precipitation of dissolved ions in water
 - *Organic* rocks produced by accumulation of biological debris, such as in swamps or bogs
- Sedimentary rock types and *sedimentary structures* within the rocks give clues to *past environments*
- *Fossils* in sedimentary rocks give clues to the history of life
- Important *resources* (coal, oil) are found in sedimentary rocks